

Summary and Highlights: Spark Architecture

In this lesson, you learned that:

- Spark architecture has driver and executor processes coordinated by the SparkContext in the driver.
- The driver creates jobs, and the Spark Context splits jobs into tasks that can be run in parallel in the executors on the cluster. Stages are a set of tasks that are separated by a data shuffle.
- Shuffles are costly, as they require data serialization, disk, and network I/O. The driver program can be run in either client mode (connecting the driver outside the cluster) or cluster mode (running the driver in the cluster).
- Cluster managers acquire resources and run as an abstracted service outside the application. Spark can run on Spark Standalone, Apache Hadoop YARN, Apache Mesos, or Kubernetes cluster managers with specific set-up requirements.
- Choosing a cluster manager depends on your data ecosystem and factors such as ease of configuration, portability, deployment, or data partitioning needs. Spark can also run using local mode, which is useful for testing or debugging an application.
- "spark-submit" is a unified interface to submit the Spark application, no matter the cluster manager or application language.
- Mandatory options include telling Spark which cluster manager to connect to; other options set driver deploy mode or executor resourcing.
- To manage dependencies, application projects or libraries must be accessible for driver and executor processes, for example, by creating a Java or Scala uber-JAR.
- Spark Shell simplifies working with data by automatically initializing the SparkContext and SparkSession variables and providing Spark API access.